

1.1.Problem Statement

Current legacy systems in the airline industry often struggle with the sheer velocity of modern web traffic, leading to critical failures. One of the most prevalent issues is the "race condition" in database management, where two users attempt to book the last available seat at the exact same millisecond. Without rigorous concurrency control, this results in double-booking, leading to customer dissatisfaction and operational headaches for airline staff. Furthermore, many existing platforms suffer from "latency spikes" during holiday seasons or promotional sales, causing the system to hang or crash entirely under pressure. Data fragmentation is another hurdle; when information is stored in silos, inconsistencies arise between the booking agent's view and the actual aircraft manifest. Security is perhaps the most pressing concern, as flight databases are prime targets for cyberattacks aimed at stealing sensitive passenger identities and payment details. Many older systems lack modern encryption standards, making them vulnerable to sophisticated breaches. There is also the issue of "orphan records," where a payment is processed but the seat reservation fails to link, creating a vacuum in data integrity. This proposal addresses these gaps by identifying the need for a synchronized, high-speed, and fortified database environment.

1.2.Objectives

The primary ambition of this project is to architect a database that serves as a "single source of truth" for all flight-related operations. A major specific objective is the implementation of strict data constraints at the schema level to make seat duplication mathematically impossible within the system logic. We also aim to achieve "zero-lag" status for availability updates, ensuring that the moment a seat is selected, it is visible as "unavailable" across the entire global network. Security is a

non-negotiable objective, focusing on the integration of end-to-end encryption for all financial transactions and personal passenger data. From a performance standpoint, the objective is to maintain a sub-second response time for 95% of all queries, even during peak load periods. We are also committed to ensuring full compliance with ACID (Atomicity, Consistency, Isolation, Durability) properties, which are the gold standard for reliable transaction processing.

Supporting high concurrency is vital, meaning the system must handle thousands of simultaneous read/write operations without compromising data quality. Additionally, the project seeks to provide a scalable framework that can easily expand from a single regional airline to a multi-national carrier network. Ultimately, these objectives converge to create a system that is resilient, transparent, and highly efficient for all stakeholders.